



RO-3-5120

5120 Bit Static Read Only Memory

FEATURES

- 512×10 Organization
- Single +5 Volt Supply
- TTL/DTL Compatible
- Static Operation—no clocks required
- 500ns Maximum Access Time
- 150mW Typical Power
- Three-State Outputs—under control of 'Output Inhibit' signal
- Totally Automated Custom Programming
- Zener Protected Inputs
- Glass Passivation Protection

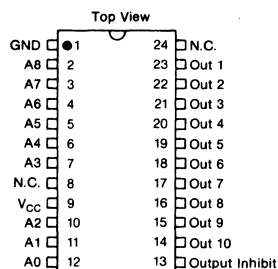
DESCRIPTION

The General Instrument RO-3-5120 is a 5120 bit static Read-Only-Memory. It is organized as 512 ten bit words and requires 9 bits of addressing. An 'Output Inhibit' function is provided to simplify the connection of several ROMs to a common bus. The RO-3-5120 is constructed on a single monolithic chip utilizing low-voltage N-channel Ion Implant technology.

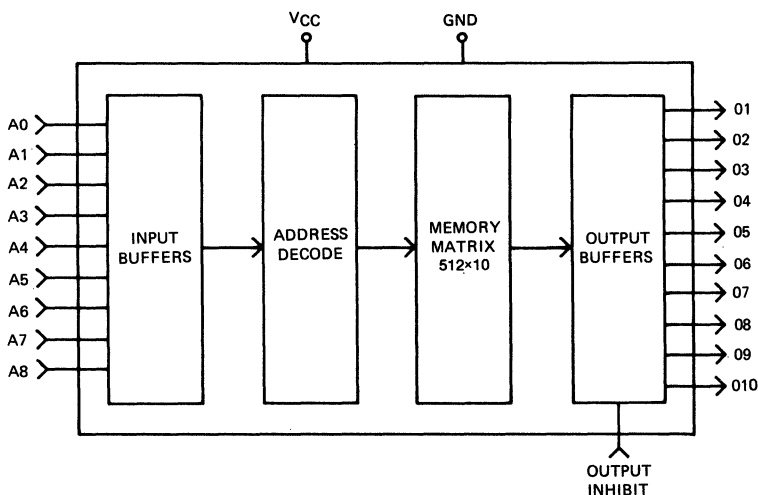
A separate publication, "RO-3-5120 Custom Coding Information," available from GI Sales Offices, describes the punched card and truth table data format for custom programming of the RO-3-5120 memory.

PIN CONFIGURATION

24 LEAD DUAL IN LINE



BLOCK DIAGRAM





ELECTRIC CHARACTERISTICS

Maximum Ratings*

V_{CC} and input voltages (with respect to GND) -0.3V to +8.0V
 Storage Temperature -65°C to +150°C
 Operating Temperature (T_A) 0°C to +70°C

*Exceeding these ratings could cause permanent damage. Functional operation of this device at these conditions is not implied—operating ranges are specified below.

Standard Conditions (unless otherwise noted)

$V_{CC} = +5$ Volts $\pm 5\%$

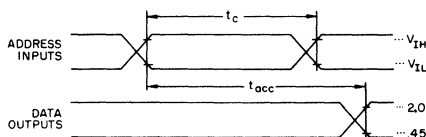
Temperature (T_A) = 0°C to +70°C

Output Loading: One TTL Load, $C_{L\text{ TOTAL}} = 50\text{pF}$.

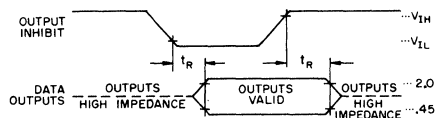
Characteristic	Sym	Min	Typ**	Max	Units	Conditions
DC CHARACTERISTICS						
Address, Output Inhibit Inputs						
Logic "1"	V_{IH}	2.2	—	—	V	
Logic "0"	V_{IL}	—	—	0.65	V	
Leakage	I_{LI}	—	—	10	μA	
Data Outputs						
Logic "1"	V_{OH}	2.2	—	—	V	$I_{OH} = 100\mu\text{A}$
Logic "0"	V_{OL}	—	—	0.45	V	$I_{OL} = 1.6\text{mA}$
Leakage	I_{LO}	—	—	10	μA	
Power Supply Current						
I_{CC}	—	—	30	45	mA	Outputs open
AC CHARACTERISTICS						
Inputs						
Cycle Time	t_c	500	—	—	ns	$f = 1\text{MHz}$
Capacitance	C_i	—	5	8	pF	
Data Outputs						
Access Time	t_{ACC}	—	350	500	ns	
Inhibit Response Time	t_R	—	—	200	ns	
Capacitance	C_o	—	8	10	pF	$f = 1\text{MHz}$

**Typical values are at +25°C and nominal voltages

TIMING DIAGRAMS



**ACCESS TIME (ADDRESS TO OUTPUT—
OUTPUT INHIBIT AT LOGIC '0')**



**INHIBIT RESPONSE TIME
(ADDRESS INPUTS STABLE)**